A-level

ENVIRONMENTAL SCIENCE

Paper 2 - Generated practice paper Set 39

Mark scheme

120 marks: Questions 1-10 (95 marks) plus one essay from Question 11 (25 marks).

Marking guidance

• Each part opens with a line in bold saying how its marks are shared out, followed by one sentence on how to mark it.
• The figure in bold at the start of a marking point is the mark that point earns. Lines indented beneath it, starting with a dash, are the detail that belongs to the same mark; a line starting **OR** is another route to it.
• **Also accept** lists other wordings and answers that earn the mark - wording never has to match these notes, only the science. **Do not credit** lists the answers that look right and earn nothing.
• Where a question asks for a fixed number of points (eg *two* reasons), credit only the first points given.
• For levels-of-response questions, determine the level from the descriptors first, then position the mark within the level using the indicative content.
• Calculations: credit correct working (method marks) even if the final answer is wrong; incorrect answers with no working score 0.

Textbook references: every question below cites the printed page numbers in R. Genn, Environmental Science A level AQA (Insight & Perspective, 2018) from which the content is drawn, together with the AQA 7447 specification reference.

Question	Marking guidance	Marks
01.1	**1 mark each** 1 mark for each of the 3 points below. **1** Plants close their stomata to reduce transpiration losses **1** Closed stomata stop the absorption of carbon dioxide so photosynthesis and growth stop **1** Less water for nutrient absorption / transport and cell turgidity – severe shortage dehydrates cells and inhibits biochemical reactions **Also accept** • Any other correct, relevant point of the same kind - wording never has to match these notes. • **transpiration** → water lost as vapour from the leaves	3

Do not write outside the box

Question	Marking guidance	Marks
01.2	**Marks shared** Up to 4 marks in total - share them between the 3 points below according to how completely each one is made. **up to 2** Anaerobic soil favours denitrifying bacteria and inhibits nitrifying bacteria so nitrate is lost faster than it forms – higher risk of fungal disease – root respiration inhibited **up to 2** Drainage ditches or pipes **up to 2** Deep ploughing – avoiding compaction by machinery / livestock – adding organic matter to encourage worms **Also accept** • Any other correct, relevant point of the same kind - wording never has to match these notes. • **waterlogged** → soil pores filled with water so there is no air • **denitrifying** → nitrate converted back to nitrogen gas • **nitrifying** → ammonium oxidised to nitrite and then nitrate	4
01.3	**1 mark each** 1 mark for each of the 2 points below. **1** Groundwater often contains dissolved salts – when the water evaporates from the soil the salts are left behind and accumulate **1** High soil salinity causes osmotic dehydration of roots / crop death – washing out with extra water uses more water and salinises rivers **Also accept** • Any other correct, relevant point of the same kind - wording never has to match these notes. • **salinisation** → salt building up in the soil / soil becoming salty	2
01.4	1 mark for the point below. **1** Add crushed lime (calcium carbonate / hydroxide) – low pH increases leaching, inhibits nutrient uptake and mobilises toxic aluminium and lead **Also accept** • Any other correct, relevant point of the same kind - wording never has to match these notes. • **leaching** → nutrients or ions washed down out of the soil / washed away by water	1
	Source: R. Genn, Environmental Science A level AQA (Insight & Perspective, 2018), pp. 308-310, 313; AQA spec 3.5.1.1.	
02.1	**Calculation** A correct final answer scores full marks on its own. If the answer is wrong, each correct step of working below earns its mark. **1** Extensive cereal 5.0 – intensive cereal 3.0 **1** Intensive beef 0.1 **Also accept** • The correct answer however it is set out, including a different but valid method. • An answer carried forward from an earlier incorrect step (error carried forward). • An answer rounded to 2 or more significant figures, and the answer written as a fraction, decimal or percentage where the question does not specify. **Do not credit** • A bare wrong answer with no working.	2

Do not write outside the box

Question	Marking guidance	Marks
02.2	**1 mark each** 1 mark for each of the 3 points below. **1** Intensive cereal is most productive per hectare (60 GJ) but less efficient (3.0) than extensive (5.0) **1** Each extra unit of input gives a smaller extra yield (law of diminishing returns) **1** Beef has very low productivity and efficiency because energy is lost at an extra trophic level **Also accept** • Any other correct, relevant point of the same kind - wording never has to match these notes. • The converse of any point, ie the same comparison written the other way round. • **trophic level** → the feeding level in a food chain **Do not credit** • 'energy is lost between trophic levels' on its own - say how it is lost: respiration, excretion, egestion and uneaten parts	3
02.3	**Marks shared** Up to 3 marks in total - share them between the 2 points below according to how completely each one is made. **up to 2** Any input that increases productivity but requires the use of energy (usually fossil fuel) **up to 2** Manufacture of nitrate fertiliser (Haber process) and pesticides – fuel for machinery – pumping irrigation water – grain drying – food processing and transport – embodied energy of machinery **Also accept** • Any other correct, relevant point of the same kind - wording never has to match these notes. • **embodied energy** → all the energy used to make and transport a product **Do not credit** • Points after the first three - the question asks for three, so a list of extras cannot pick up the marks. • pests 'getting used to' a pesticide - the few naturally resistant individuals survive and pass the trait on	3
02.4	**1 mark each** 1 mark for each of the 2 points below. **1** Because of diminishing returns, the first units of input on unfertilised land give the largest yield increases **1** Moving inputs from intensively farmed land (where extra input adds little) to extensively farmed land gives a greater total yield - though land, cost and distribution limit this **Also accept** • Any other correct, relevant point of the same kind - wording never has to match these notes.	2
	Source: R. Genn, Environmental Science A level AQA (Insight & Perspective, 2018), pp. 324-327; AQA spec 3.5.1.2.	

Do not write outside the box

Question	Marking guidance	Marks
03.1	**1 mark each** 1 mark for each of the 3 points below. **1** Readings drift so they may not be accurate – calibration adjusts the meter to match a known standard so results are accurate and comparable between meters / occasions **1** pH meter placed in buffer solutions of known pH (eg 4, 7, 10) and adjusted to read correctly **1** If a meter cannot be adjusted, record the percentage error and correct the results **Also accept** • Any other correct, relevant point of the same kind - wording never has to match these notes. • **accurate** → close to the true value	3
03.2	**1 mark each** 1 mark for each of the 3 points below. **1** Same height above the ground **1** No obstacles to air flow – impeller type facing into the wind **1** Axis of rotation horizontal (impeller) or vertical (cup) – same time period / averaging **Also accept** • Any other correct, relevant point of the same kind - wording never has to match these notes.	3
03.3	**1 mark each** 1 mark for each of the 2 points below. **1** Two thermometers, one with a wet bulb, are spun until the wet bulb stops cooling **1** Evaporation cools the wet bulb more when humidity is low – the temperature difference is compared with a table to find humidity **Also accept** • Any other correct, relevant point of the same kind - wording never has to match these notes.	2
03.4	**Any 2 of 3** 1 mark each for any 2 of the 3 points below, in any order, maximum 2. **1** Test strips with reagent bands compared with a colour chart (semi-quantitative) **1 Electronic colourimeter:** reagent added and colour intensity measured at a specific wavelength **1** Ion-selective electrode measuring electrode potential (other ions can interfere) **Also accept** • Any other correct, relevant point of the same kind - wording never has to match these notes. **Do not credit** • Points after the first two - the question asks for two, so a list of extras cannot pick up the marks.	2
	Source: R. Genn, Environmental Science A level AQA (Insight & Perspective, 2018), pp. 407-410, 413; AQA spec 3.7.2.1.	

Do not write outside the box

Question	Marking guidance	Marks
04.1	**2 marks per point** 2 marks for each of the 3 points below; award 1 mark where a point is only partly made. **2** Photosynthesis and aerobic respiration balance O2 and CO2 in a dynamic equilibrium – evapotranspiration adds water vapour forming clouds and rain – microbes drive the nitrogen and other biogeochemical cycles so wastes do not accumulate and nutrients are recycled **2** Detritivores, fungi and bacteria decompose dead organic matter releasing nutrients and forming humus that binds soil – organic acids weather rock forming new soil – roots bind soil, reducing erosion and regulating water flow / flooding **2** Species provide food, pollination (bees, moths, birds, bats - more reliable than wind, especially in forests), seed dispersal (animals carry larger seeds to suitable sites, faeces as fertiliser, gut passage stimulates germination) and habitat provision (trees as nest sites and canopy microclimate; mollusc shells for hermit crabs) **Also accept** • Any other correct, relevant point of the same kind - wording never has to match these notes. • **evapotranspiration** → evaporation plus transpiration • **humus** → dark, decomposed organic matter in soil • **reliable** → repeatable / gives consistent results when repeated **Do not credit** • 'repeat the readings to make them more accurate' - repeats improve reliability; accuracy depends on the method and the instrument	6
04.2	**1 mark each** 1 mark for each of the 2 points below. **1** Its nectar is at the end of a 30 cm tube reached only by the long tongue of the sphinx moth, which carries pollen only between orchids of the same species **1 Advantage:** no pollen wasted on other species and the moth has an exclusive food source – risk: if either species declines the other cannot survive **Also accept** • Any other correct, relevant point of the same kind - wording never has to match these notes.	2
04.3	**1 mark each** 1 mark for each of the 2 points below. **1** They are not ruminants so seeds are not destroyed by repeated chewing / pass through intact – they eat the fruit of most tree species **1** Not territorial so they carry seeds over large distances / areas **Also accept** • Any other correct, relevant point of the same kind - wording never has to match these notes.	2
	Source: R. Genn, Environmental Science A level AQA (Insight & Perspective, 2018), pp. 18-19; AQA spec 3.1.2.1.3.	

Do not write outside the box

Question	Marking guidance	Marks
05.1	**1 mark each** 1 mark for each of the 3 points below. **1** The range of an environmental factor within which a species / population can survive (survival highest at the optimum, falling towards the limits) **1** X survives from about pH 5.5 to 8.5 (wide range) – Y only from about pH 6.0 to 8.0 (narrow range) **1** Both have the same optimum (pH 7.0) **Also accept** • Any other correct, relevant point of the same kind - wording never has to match these notes. • The converse of any point, ie the same comparison written the other way round. • **range of tolerance** → the range of conditions in which a species can survive	3
05.2	**1 mark each** 1 mark for each of the 3 points below. **1** X: 60% survive at pH 6.0 whereas only 5% of Y survive **1** X has a larger gene pool / wider range of tolerance so some individuals are adapted to the new conditions **1** Survivors breed and their offspring are adapted so the population rebuilds – Y is likely to die out locally **Also accept** • Any other correct, relevant point of the same kind - wording never has to match these notes. • **gene pool** → the range of alleles / genes present in a population • **range of tolerance** → the range of conditions in which a species can survive	3
05.3	**1 mark each** 1 mark for each of the 2 points below. **1** Reveals the habitat requirements / which abiotic factor is near the edge of tolerance **1** Management can make small beneficial changes to that factor (eg liming, shading, water levels) to keep conditions within the range and increase survival **Also accept** • Any other correct, relevant point of the same kind - wording never has to match these notes. • **range of tolerance** → the range of conditions in which a species can survive • **abiotic** → non-living / physical factors	2
05.4	**Any 2 of 5** 1 mark each for any 2 of the 5 points below, in any order, maximum 2. **1** Food supply **1** Pollinators **1** Seed dispersal species **1** Disease / pathogens **1** Predators / competitors **Also accept** • Any other correct, relevant point of the same kind - wording never has to match these notes. • **biotic** → living factors / effects of other organisms **Do not credit** • Points after the first two - the question asks for two, so a list of extras cannot pick up the marks.	2

Do not write outside the box

Question	Marking guidance	Marks
	Source: R. Genn, Environmental Science A level AQA (Insight & Perspective, 2018), pp. 72-73; AQA spec 3.1.3.1.	
06.1	**1 mark each** 1 mark for each of the 4 points below. **1** No major temperature extremes – regular water supply with no pronounced dry season – deep fertile soils built by leaf litter and root binding – strong seasonality **1** Deciduous canopy present spring to autumn shades the floor during the best growing temperatures **1** Bluebells, snowdrops, lesser celandine grow and flower early in spring before the trees leaf **1** Winter food shortage causes animals to hibernate (dormice), store food or migrate **Also accept** • Any other correct, relevant point of the same kind - wording never has to match these notes.	4
06.2	**1 mark each** 1 mark for each of the 3 points below. **1** Long time for species to colonise – some woodland plants colonise very slowly so their presence (ancient woodland indicators) shows the wood is old **1** Natural woodland has trees of all ages with clearings where old trees died, giving varied light and dead wood habitats **1** A new plantation has a single-age dense canopy with low light and few niches until trees start to die (over 200 years) **Also accept** • Any other correct, relevant point of the same kind - wording never has to match these notes. • **biodiversity** → the variety of life: habitat, species and genetic diversity	3
06.3	**Any 3 of 4** 1 mark each for any 3 of the 4 points below, in any order, maximum 3. **1** Clearance for farmland, plantations, urban development, roads, mineral extraction **1** Fragmentation isolating remaining woods so populations cannot move between them and are vulnerable **1 Management change:** end of coppicing so woods become uniform and dark – replacement by conifer monocultures **1** Overgrazing by deer preventing regeneration – introduced pathogens (ash dieback) **Also accept** • Any other correct, relevant point of the same kind - wording never has to match these notes. **Do not credit** • Points after the first three - the question asks for three, so a list of extras cannot pick up the marks.	3
	Source: R. Genn, Environmental Science A level AQA (Insight & Perspective, 2018), pp. 50-52; AQA spec 3.1.2.3.4.	

Do not write outside the box

Question	Marking guidance	Marks
07.1	**1 mark each** 1 mark for each of the 3 points below. **1** Interconnected biological processes recycle nutrient elements so small amounts support life indefinitely without depletion – waste products of one process are raw materials of another so wastes do not build up **1** Transpiration returned water vapour to the atmosphere increasing rainfall elsewhere **1** Allowing more plant growth, further stabilising climate and soils **Also accept** • Any other correct, relevant point of the same kind - wording never has to match these notes. • **transpiration** → water lost as vapour from the leaves	3
07.2	**Any 3 of 4** 1 mark each for any 3 of the 4 points below, in any order, maximum 3. **1** Lack of ancient historical / instrumental data **1** Reliance on proxy data (tree rings, pollen) of limited reliability for ancient conditions **1** Lack of sophisticated equipment / inability to measure many factors **1** Limited coordination between researchers – lack of data collection in many areas (oceans, poles) **Also accept** • Any other correct, relevant point of the same kind - wording never has to match these notes. • **reliability** → repeatable / gives consistent results when repeated **Do not credit** • Points after the first three - the question asks for three, so a list of extras cannot pick up the marks. • 'repeat the readings to make them more accurate' - repeats improve reliability; accuracy depends on the method and the instrument	3
07.3	**1 mark each** 1 mark for each of the 4 points below. **1** The ratio of oxygen-18 to oxygen-16 in the ice indicates the temperature when the snow fell **1** Air bubbles trapped in ice give the composition of the past atmosphere (CO_2, methane) **1** Ratio of carbon-14 to carbon-12 dates organic material (eg pollen in sediments) **1** More magnesium is incorporated into calcite at higher temperatures, so the ratio estimates past sea temperature **Also accept** • Any other correct, relevant point of the same kind - wording never has to match these notes.	4
	Source: R. Genn, Environmental Science A level AQA (Insight & Perspective, 2018), pp. 8-9; AQA spec 3.1.1.2-3.1.1.3.	
08.1	**Correct answer only** 1 mark for the correct option. No mark if more than one box is ticked. **1** B **Also accept** • **abiotic** → non-living / physical factors	1

Do not write outside the box

Question	Marking guidance	Marks
08.2	**Any 4 of 5** 1 mark each for any 4 of the 5 points below, in any order, maximum 4. **1** Deforestation **1** Ploughing of grassland / expansion of farmland **1** Reservoir creation / flooding **1** Mineral extraction, especially open-cast mining **1** Urban expansion **Also accept** • Any other correct, relevant point of the same kind - wording never has to match these notes. **Do not credit** • Points after the first four - the question asks for four, so a list of extras cannot pick up the marks. • 'trees give us the oxygen we breathe' as the main impact - credit carbon storage, soil protection, the water cycle and habitat loss • a dam described as having no environmental impacts - flooded land, blocked fish migration, trapped sediment and methane from drowned vegetation	4
	Source: R. Genn, Environmental Science A level AQA (Insight & Perspective, 2018), pp. 20-26; AQA spec 3.1.2.2.	
09.1	**1 mark each** 1 mark for each of the 4 points below. **1** In winter nutrients are high (storms disturb sediments, deep mixing) but light is too low for photosynthesis **1** In spring light rises while nutrients are still high so growth is rapid - the spring bloom (peak about May) **1** The bloom uses up nutrients (falling to 10 by July) – dead plankton sink out of the photic zone carrying nutrients to the seabed **1** So nutrients become limiting in summer despite high light – a smaller autumn bloom follows when storms return nutrients while light is still adequate **Also accept** • Any other correct, relevant point of the same kind - wording never has to match these notes.	4
09.2	**1 mark each** 1 mark for each of the 2 points below. **1** Warm surface water is stable (no seasonal mixing / storms) so nutrients lost to deep water in sinking dead organisms are not replaced **1** Nutrient (especially phosphate) concentrations in the photic zone remain very low - the limiting factor **Also accept** • Any other correct, relevant point of the same kind - wording never has to match these notes. • **limiting factor** → the factor in shortest supply that holds the rate or population back	2

Do not write outside the box

Question	Marking guidance	Marks
09.3	**1 mark each** 1 mark for each of the 3 points below. **1** Upwelling (offshore winds, seamounts, currents deflected upwards) brings cold nutrient-rich deep water into the photic zone **1** Coastal waters receive nutrients washed off the land in rivers and storms disturb shallow sediments **1** Nutrients plus light in the shallow photic zone allow high algal productivity supporting rich food webs and fisheries (eg Peru anchovy) **Also accept** • Any other correct, relevant point of the same kind - wording never has to match these notes. **Do not credit** • 'the algae use up all the oxygen' on its own - the oxygen is used by aerobic decomposers breaking down the dead algae	3
09.4	1 mark for the point below. **1** Phosphates have low solubility and no atmospheric reservoir – they are only added by river runoff or upwelling and are lost to sediments **Also accept** • Any other correct, relevant point of the same kind - wording never has to match these notes. **Do not credit** • a dam described as having no environmental impacts - flooded land, blocked fish migration, trapped sediment and methane from drowned vegetation	1
	Source: R. Genn, Environmental Science A level AQA (Insight & Perspective, 2018), pp. 338-339; AQA spec 3.5.2.1.	

Do not write outside the box

Question	Marking guidance	Marks
10.1	**Levels of response** Decide the level from the descriptors first, then use the indicative content to place the mark within that level. The points below are not a checklist. **Level 3 (7-9 marks)**: Answer is comprehensive, relevant and logically structured. Shows detailed knowledge and understanding of the relevant scientific ideas and applies them accurately to the context. Evaluation/analysis is well supported by evidence and a clear judgement or conclusion is reached. **Level 2 (4-6 marks)**: Answer is mostly relevant and shows some structure. Knowledge and understanding of relevant ideas is sound but application to the context or evaluation may be incomplete or lack detail. A judgement may be made but is not fully supported. **Level 1 (1-3 marks)**: Answer is basic, with limited structure. Some relevant knowledge is shown but there are omissions or errors. Little application to the context and little or no evaluation. **0 marks**: No relevant content. **Indicative content - a guide to placing the mark, not a checklist** • **Context:** wild catches have not increased since about 1990 as stocks are fully or over-exploited – aquaculture output growing rapidly and approaching the wild catch – fish are the last large-scale hunted food • **Advantages of aquaculture:** control of limiting factors, selective breeding, gender control, high yields per area – fish have low FCRs (no thermoregulation / support) compared with livestock – herbivores (tilapia, carp) and molluscs very efficient – polyculture, IMTA, aquaponics, rice-fish • **Limits:** carnivorous species (salmon, shrimp) need fishmeal from wild fish so still depend on fishing and deplete forage fish (sandeels - puffins) – some species (bluefin tuna) will not breed in captivity – consumer preference for carnivores • **Environmental impacts:** habitat loss (mangroves), organic pollution, pesticides, antibiotics and resistance, lice, escapes and gene pool contamination, predator control – and how they can be reduced • Fishing can be sustainable with quotas, gear design, NTZs, size limits, effort limits – MSY management needs good data • **Judgement:** aquaculture can supply much of the growth in demand, especially herbivores and shellfish, but cannot fully replace fishing – both need sustainable management **Also accept** • Any other correct, relevant point of the same kind - wording never has to match these notes. • Any accurate, relevant material that is not in the indicative content. • **aquaculture** → fish or shellfish farming	9
	• **gene pool** → the range of alleles / genes present in a population • **sustainable** → meeting present needs without stopping future generations meeting theirs **Do not credit** • pests 'getting used to' a pesticide - the few naturally resistant individuals survive and pass the trait on • 'sustainable means it lasts a long time' - it means use that can continue without reducing what is available in future	
	Model answer - Level 3 (9 marks)	

Do not write outside the box

Question	Marking guidance	Marks
	Wild catches have not increased since about 1990 because most stocks are fully or over-exploited, while aquaculture has grown rapidly and now approaches the wild catch, so the question is how far it can take over. Its advantages are real: limiting factors, breeding, gender and disease can be controlled, yields per area are high, and fish have low food conversion ratios because they need not support their weight or keep warm; herbivores such as tilapia and carp and filter-feeding molluscs need no fishmeal at all.	
	Its limits are equally real. The carnivorous species consumers prefer - salmon, shrimp - are fed fishmeal from wild forage fish, so their farming still depends on fishing and depletes species such as sandeels on which puffins rely; some species, such as bluefin tuna, will not breed in captivity. Farms destroy mangroves, release organic waste, pesticides and antibiotics that breed resistance, spread lice, allow escapes that contaminate wild gene pools and provoke predator control, though siting, closed systems, polyculture and vaccines can reduce these. Meanwhile fishing itself can be sustainable if quotas based on good stock data, gear design, size and effort limits and no-take zones are enforced.	
	Aquaculture can therefore supply most of the growth in demand, especially through herbivorous fish and shellfish, and can relieve pressure on wild stocks, but it cannot fully replace fishing while carnivorous species depend on wild fishmeal and some species cannot be farmed. Both must be managed sustainably; neither is a substitute for the other.	
	Weaker answer - Level 1 (2 marks)	
	Aquaculture is farming fish instead of catching them. Fish farms can produce a lot of fish and they do not run out like wild fish do. But fish farms cause pollution and diseases can spread to wild fish. Salmon farms feed the salmon on fish which have to be caught. So aquaculture can replace some fishing but not all of it.	
	What makes the difference	
	• Level 1 reaches a reasonable conclusion but supports it with two sentences; Level 3 gives the context (flat wild catch since 1990), the advantages in detail and the impacts with ways to reduce them.	
	• Level 3 distinguishes between species types (herbivores and molluscs vs carnivores) - the key to 'to what extent'.	
	• Level 3 also considers whether fishing can be made sustainable, which the comparison requires.	
10.2	1 mark for the point below. **1** A species that is given artificial food (pellets) rather than relying on natural food in the water, eg salmon, shrimp **Also accept** • Any other correct, relevant point of the same kind - wording never has to match these notes. • **aquaculture** → fish or shellfish farming	1
	Source: R. Genn, Environmental Science A level AQA (Insight & Perspective, 2018), pp. 338-359; AQA spec 3.5.2.	

Do not write outside the box

Question	Marking guidance	Marks
11.1	**Discuss the extent to which agriculture can be made sustainable while feeding a growing human population.** **Levels of response** **Level 5 (21-25 marks)**: Comprehensive and detailed knowledge and understanding drawn from across the specification, applied accurately to the question. Argument is coherent, well-structured and supported by a wide range of relevant, well-chosen examples and quantitative detail where appropriate. Interconnections between topics are made explicitly and a clear, justified conclusion is reached. **Level 4 (16-20 marks)**: Detailed knowledge and understanding with good application to the question. Argument is logically organised with relevant examples; some links between different areas of the specification are made. Conclusions are drawn but may be less fully justified. **Level 3 (11-15 marks)**: Sound knowledge and understanding, mostly applied to the question. Some structure and relevant examples, but coverage may be uneven or descriptive rather than analytical. Limited linking of topics. **Level 2 (6-10 marks)**: Basic knowledge and understanding with partial relevance. Answer is largely descriptive with few examples and limited structure; some errors or omissions. **Level 1 (1-5 marks)**: Limited, fragmented knowledge with little relevance to the question. Little structure; examples absent or inappropriate. **0 marks**: No relevant content. **Indicative content - students are not expected to cover all of these** • **Need:** population possibly 12 billion – humans raise carrying capacity – trophic level of diet • **Impacts to be reduced:** habitat loss, drainage, nutrient enrichment, introduced species, pesticide and nutrient pollution, nitrate toxicity, greenhouse gases, hydrological changes, soil erosion • **Social, economic and political influences:** consumer choices, technology availability, grants, guaranteed markets, surpluses, quotas, set-aside, agri-environment schemes • **Strategies:** cultural and biological pest control, integrated control, organic nutrient supply, legumes, low tillage, water conservation, CWR conservation, renewable energy, methane reduction, habitat retention • **Limits:** little new land, imports reducing local availability, cost and acceptance of new techniques, past land degradation • Justified conclusion *Credit any relevant, accurate material, including points that are not listed here. Reward a clear line of argument and a supported conclusion.* **Source:** R. Genn, Environmental Science A level AQA (Insight & Perspective, 2018), pp. 328-337; AQA spec 3.5.1.3-3.5.1.5.	25

Do not write outside the box

Question	Marking guidance	Marks
11.2	**Discuss how human activities influence biodiversity, using examples from a range of different contexts.** **Levels of response** **Level 5 (21-25 marks)**: Comprehensive and detailed knowledge and understanding drawn from across the specification, applied accurately to the question. Argument is coherent, well-structured and supported by a wide range of relevant, well-chosen examples and quantitative detail where appropriate. Interconnections between topics are made explicitly and a clear, justified conclusion is reached. **Level 4 (16-20 marks)**: Detailed knowledge and understanding with good application to the question. Argument is logically organised with relevant examples; some links between different areas of the specification are made. Conclusions are drawn but may be less fully justified. **Level 3 (11-15 marks)**: Sound knowledge and understanding, mostly applied to the question. Some structure and relevant examples, but coverage may be uneven or descriptive rather than analytical. Limited linking of topics. **Level 2 (6-10 marks)**: Basic knowledge and understanding with partial relevance. Answer is largely descriptive with few examples and limited structure; some errors or omissions. **Level 1 (1-5 marks)**: Limited, fragmented knowledge with little relevance to the question. Little structure; examples absent or inappropriate. **0 marks**: No relevant content. **Indicative content - students are not expected to cover all of these** • **Direct exploitation:** food (fisheries, bushmeat), fashion, entertainment / pets, furniture and ornaments, traditional medicines, other products (whale oil) – deliberate eradication of predators, competitors, vectors, pests • **Changes in abiotic factors:** water availability (drainage, abstraction, HEP), light (forest clearance), oxygen (organic pollution, thermal), nutrients (eutrophication), pH (acid mine drainage, acid rain), temperature (thermal pollution, climate change), turbidity, physical damage – inactivity (stopping plagioclimax management) • **Changes in biotic factors:** introduced competitors, predators, pathogens, hybridisation – loss of pollinators, seed dispersers, keystone species – food chain impacts (sea otters, puffins, jellyfish) • **Habitat destruction:** deforestation, farmland expansion, urbanisation, mining, reservoirs – fragmentation and edge effects • **Contexts:** temperate woodland, rainforest, coral reefs, oceanic islands, mangroves, Antarctica	25
	– agriculture, fishing, forestry • **Positive influences:** habitat creation, conservation management, reintroductions, legislation • Evaluation of which influences are most damaging and hardest to reverse – conclusion *Credit any relevant, accurate material, including points that are not listed here. Reward a clear line of argument and a supported conclusion.* **Source:** R. Genn, Environmental Science A level AQA (Insight & Perspective, 2018), pp. 20-26, 328-330, 341-344, 367-369; AQA spec 3.1.2.2, 3.5.1.3, 3.5.2.2, 3.5.3.3.	

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